



FAISS Product Matching System

Accurate Metrics • Real FAISS • Comparison Charts • Fixed Calculations

 [Search Products](#)  [Benchmark Analysis](#)  [Documentation](#)

 Upload CSV Dataset



Drag and drop file here
Limit 200MB per file • CSV

Browse files



Message Group - Product.csv 0.7MB



 Dataset Preview

	S.No	BrandName	Product ID	Product Name	Brand Desc	Product Size	Currency	MRP	SellPrice	Dis
0	1	4711	FR001	Cologne Fragrance	ekw eau de cologne 400 ml	Small	Rs.	3900	3120	20
1	2	109f	DRW1	DRW1 - Westernwear-Women	womens v- neck short dress - yellow	Size:Medium,Small,X-Large,XX-Large	Rs.	1899	569	70
2	3	109f	DRW2	DRW2 - Westernwear-Women	womens round neck solid top - black	Size:Large,Medium,Small,X-Large	Rs.	1499	599	60
3	4	109f	DRW3	DRW3 - Westernwear-Women	womens round neck stripe shift dress - red	Size:Medium,Small	Rs.	1599	639	60
4	5	109f	DRW4	DRW4 - Westernwear-Women	womens round neck solid high low top - black	Size:Large,Medium,Small,X-Large	Rs.	1199	479	60
5	6	109f	DRW5	DRW5 - Westernwear-Women	womens v-neck solid top - black	Size:Large,Medium,Small,X-Small	Rs.	1199	479	60
6	7	109f	DRW6	DRW6 - Westernwear-Women	womens off shoulder neck checked top - navy	Size:Small,X-Large	Rs.	1299	519	60



Clear dataset & start over

 **Tips:** Enter a search query → select index type → choose comparison → click Search

 Search Query

men cloths

Top K Results

5

 Primary Index Type

IVF

 Compare With

FlatL2



Search Products

 Found 5 products | Recall: 0.600 | Precision: 0.600

IVF Performance

BUILD TIME

0.115s

SEARCH TIME

2.80ms

TOTAL TIME

0.118s

Recall@5

0.600

Precision@5

0.600

F1 Score

0.600

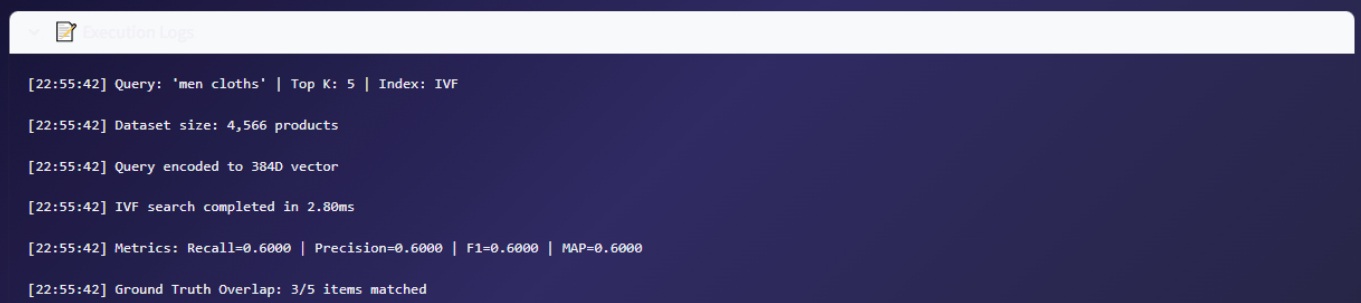
MAP@5

0.600

Matched Products

Rank #2	35.6%
S.NO	1232
BRANDNAME	and
PRODUCT ID	AND49
PRODUCT NAME	AND49 - Westernwear-Women
BRAND DESC	jacquard polyester round neck womens work wear jacket - black mix
PRODUCT SIZE	Size:8,10,14
CUSTOMER	De

Rank #4	35.3%
S.NO	1587
BRANDNAME	and
PRODUCT ID	AND404
PRODUCT NAME	AND404 - Westernwear-Women
BRAND DESC	self design round neck nylon womens party dress - black
PRODUCT SIZE	Size:8,10,18
CURRENCY	Rs





FAISS Product Matching System ↗

Accurate Metrics • Real FAISS • Comparison Charts • Fixed Calculations



Search Products



Benchmark Analysis



Documentation



Comprehensive Benchmark Testing

This will test FlatL2 vs IVF performance across multiple dataset sizes (4.5K → 200K products) and evaluate different IVF configurations with accurate Recall and Precision metrics.

Run Full Benchmark

Benchmark completed



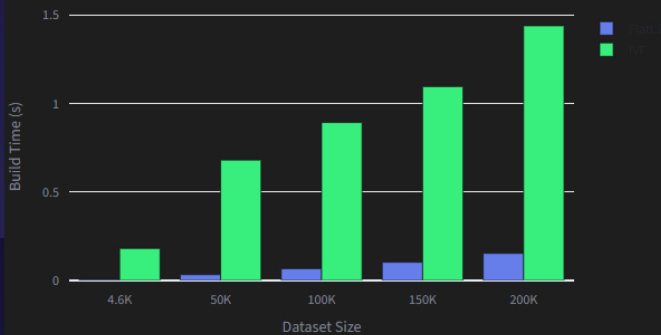
Benchmark Complete — Real FAISS Results

Max IVF Speedup: 14.84x faster than FlatL2

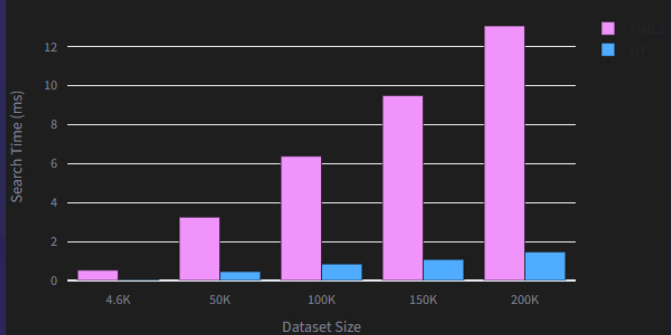
Best IVF Configuration: nlist=100, nprobe=10

Best Recall@10: 1.0000 (100.00%)

Index Build Time: FlatL2 vs IVF



Search Time per Query: FlatL2 vs IVF



Performance Comparison: FlatL2 vs IVF

Dataset Size	FlatL2 Build (s)	FlatL2 Search (ms)	IVF Build (s)	IVF Search (ms)	IVF Speedup
4.6K	0.005	0.53	0.181	0.04	14.84x
50K	0.033	3.25	0.68	0.46	7.09x
100K	0.067	6.37	0.892	0.85	7.47x
150K	0.103	9.48	1.095	1.08	8.74x
200K	0.153	13.05	1.438	1.47	8.86x

IVF Configuration Analysis

nlist	nprobe	Build Time (s)	Search Time (ms)	Recall@10	Precision@10
100	10	0.19	0.05	1	1
200	10	0.356	0.03	1	1
100	20	0.188	0.02	1	1



FAISS Product Matching System

Accurate Metrics • Real FAISS • Comparison Charts • Fixed Calculations

 Search Products  Benchmark Analysis  Documentation

FAISS Product Matching System

Key Features

Feature	Description
CSV Upload	Fast upload and processing with optimized batch encoding
Real FAISS Search	Actual FAISS indices (FlatL2, FlatIP, IVF, HNSW)
Accurate Metrics	Correct Recall@K, Precision@K, F1 Score, MAP calculations
Visual Comparison	Interactive charts comparing selected indices
Performance Analysis	Build time and search time comparisons
IVF Tuning	Test different nlist/nprobe configurations

How to Use

- Step 1: Upload Dataset** — Go to Search Products tab, upload your CSV, wait for processing.
- Step 2: Search Products** — Enter a search query, adjust Top K, select index type & comparison, click Search.
- Step 3: Analyze Results** — Review Recall, Precision, F1, MAP, and the performance comparison chart.
- Step 4: Run Benchmark** (optional) — Go to Benchmark Analysis tab and test across dataset sizes.

CSV Format Requirements

```
product_id,name,category,price,discount
1,Wireless Headphones,Electronics,99.99,15%
2,Red Cotton Shirt,Clothing,29.99,15%
3,Gaming Laptop 16GB,Computers,1299.99,5%
```

- **Best practice:** include a `name` or `product_name` column
- **Alternative:** any text columns will be combined
- **Performance:** works best with 1K–200K products

Index Types Explained

Index	Description	Best For	Speed	Accuracy
FlatL2	Exact L2 distance search	Small datasets, highest accuracy	Slow	100%
FlatIP	Exact cosine similarity	Small datasets, semantic search	Slow	100%
IVF	Inverted File Index	Large datasets, fast approximate	Fast	95–99%
HNSW	Hierarchical graph	Balanced speed/accuracy	Very Fast	97–99%

Metrics Explained

- **Recall@K**: fraction of ground-truth items retrieved — $\frac{(\text{Retrieved} \cap \text{Ground Truth})}{K}$
- **Precision@K**: fraction of retrieved items that are correct — $\frac{(\text{Retrieved} \cap \text{Ground Truth})}{\text{Retrieved}}$
- **F1 Score**: harmonic mean of Precision and Recall
- **MAP@K**: Mean Average Precision — considers ranking order of results

Performance Tips

- **Small datasets** (< 10K): use FlatL2 or FlatIP
- **Medium datasets** (10K–100K): use IVF or HNSW
- **Large datasets** (> 100K): use IVF with optimized config
- **Best accuracy**: FlatIP (exact cosine similarity)
- **Best speed**: HNSW (very fast approximate)
- **Balanced**: IVF with `nlist=200`, `nprobe=20`